Machine Learning

AIML CZG565/ DSECLZG565

M6 : Instance Based Learning

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innovate achieve lead

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Contact Session 9

Course Plan

M1	Introduction
M2	Machine learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Work Integrated Learning Programmes

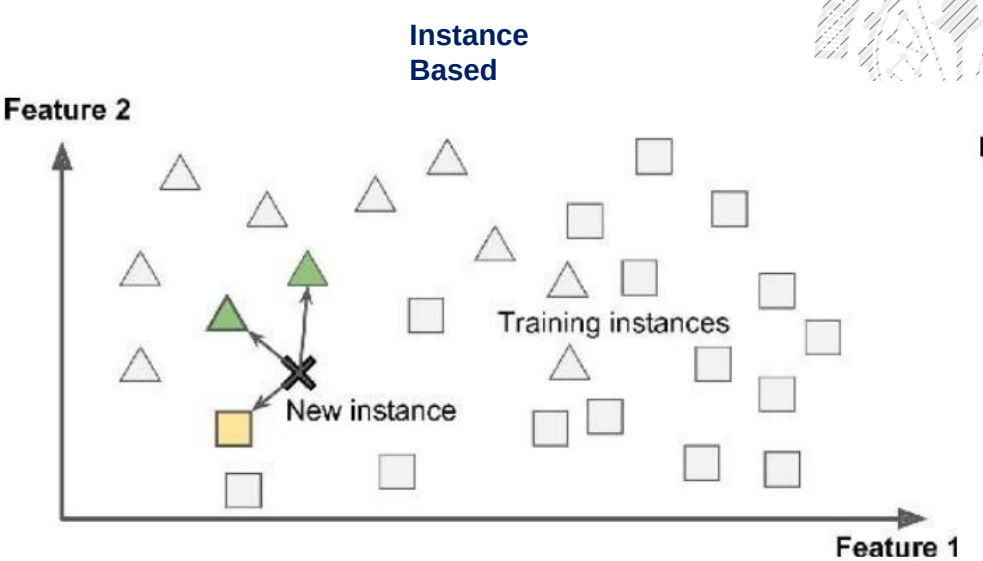
Instance Based Learning

Work Integrated Learning Programmes

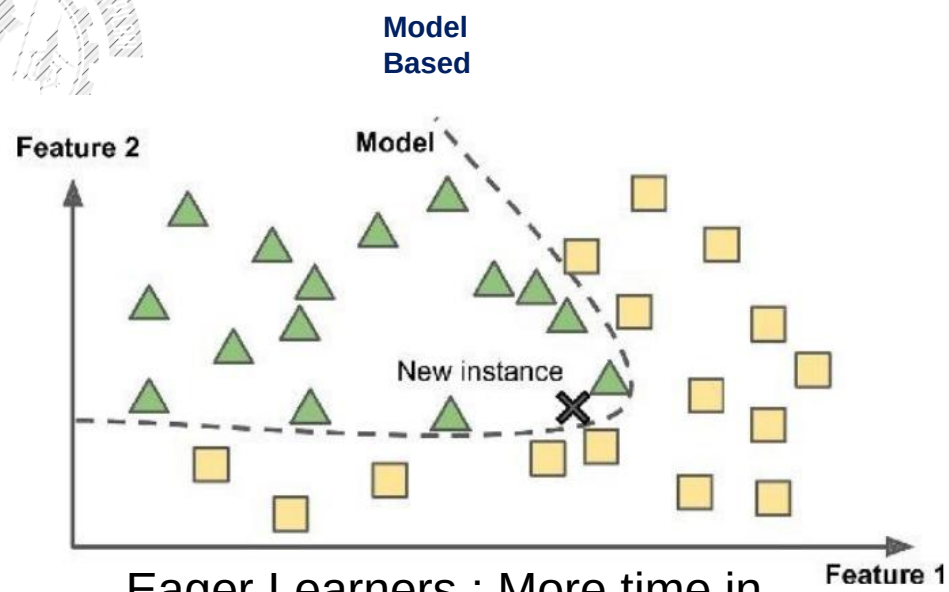
Introduction to Machine Learning

Types: Based on how & when the training data is used

- Instance Based Learning: Compare new data points to known data points
- Model Based learning : Detect patterns in the training data and build a predictive model



Lazy Learners : Less time in training but more time in predicting



Eager Learners : More time in training but Less time in predicting

Instance Based Learning

Model-based learning techniques

Use the input data

$$\begin{bmatrix} x_{1,0} & x_{1,1} & \dots & x_{1,n} \\ x_{2,0} & x_{2,1} & \dots & x_{2,n} \\ \vdots & \vdots & \vdots & \vdots \\ x_{m,0} & x_{m,1} & \cdots & x_{m,n} \end{bmatrix} \text{ and } \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{bmatrix}$$

To learn a set of parameters

$$\begin{bmatrix} \theta_0 & \theta_1 & \cdots & \theta_n \end{bmatrix}$$

Which yield a **generalized** function

$$f(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

Capable of predicting values or classes on new input data

$$f(x_i) = 39$$

$$f(x_j) = 1$$

Instance-based learning techniques

Store the input data

$$\begin{bmatrix} x_{1,0} & x_{1,1} & \dots & x_{1,n} \\ x_{2,0} & x_{2,1} & \dots & x_{2,n} \\ \vdots & \vdots & \vdots & \vdots \\ x_{m,0} & x_{m,1} & \cdots & x_{m,n} \end{bmatrix} \text{ and } \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{bmatrix}$$

When asked to predict a new value (a query)

$$y_i = ?$$

Search for similar data points previously stored

$$\begin{bmatrix} x_{4,1} & x_{4,2} & \cdots & x_{4,n} \\ x_{9,1} & x_{9,1} & \cdots & x_{9,n} \\ x_{15,1} & x_{15,1} & \cdots & x_{15,n} \end{bmatrix} \text{ and } \begin{bmatrix} y_4 \\ y_9 \\ y_{15} \end{bmatrix}$$

And use them to generate your prediction

$$y_i = \frac{y_4 + y_9 + y_{15}}{3}$$

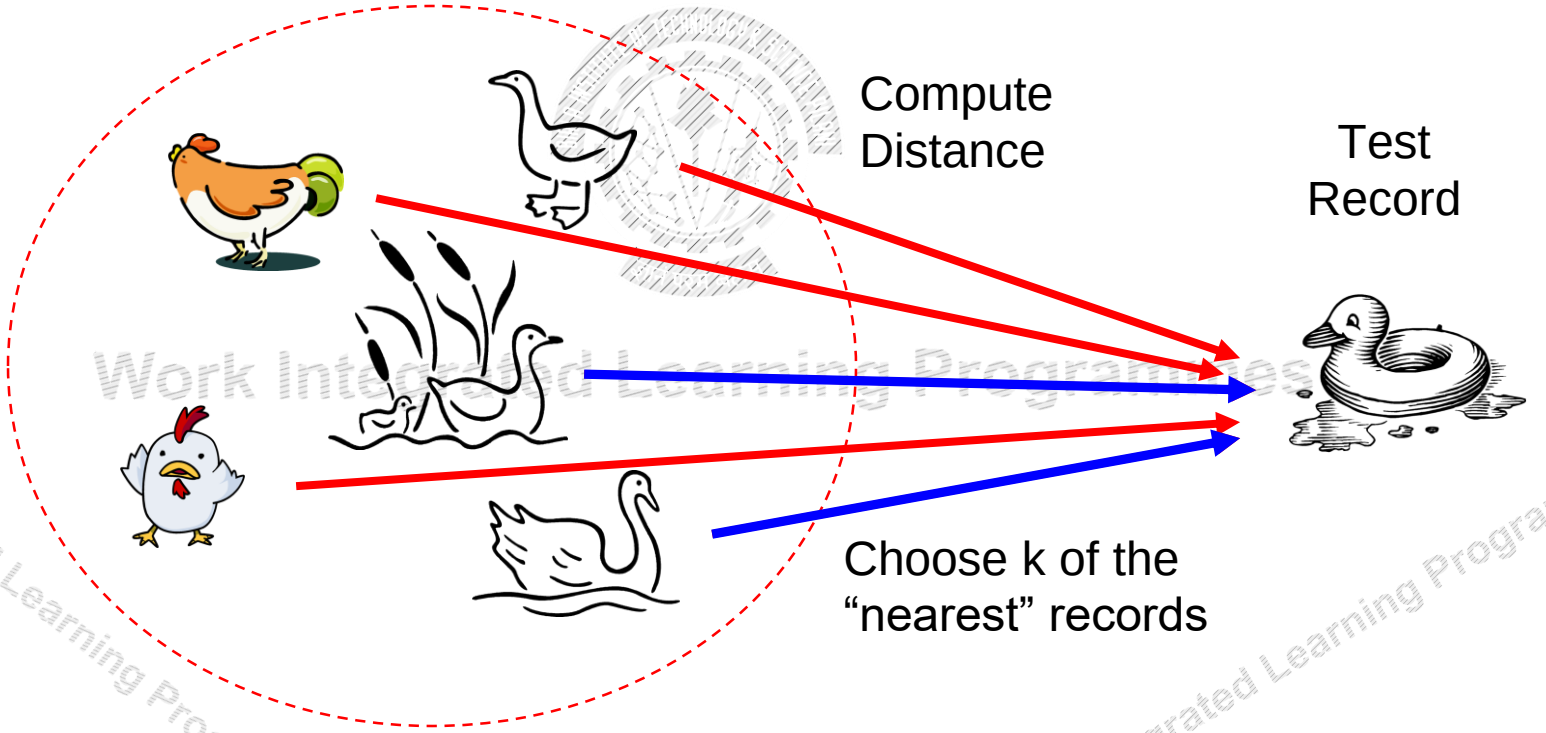
Source Credit : https://www.jeremyjordan.me/k-nearest-neighbors/

Instance Based Learning

Nearest Neighbor Approach

Basic idea:

– If it walks like a duck, quacks like a duck, then it's probably a duck

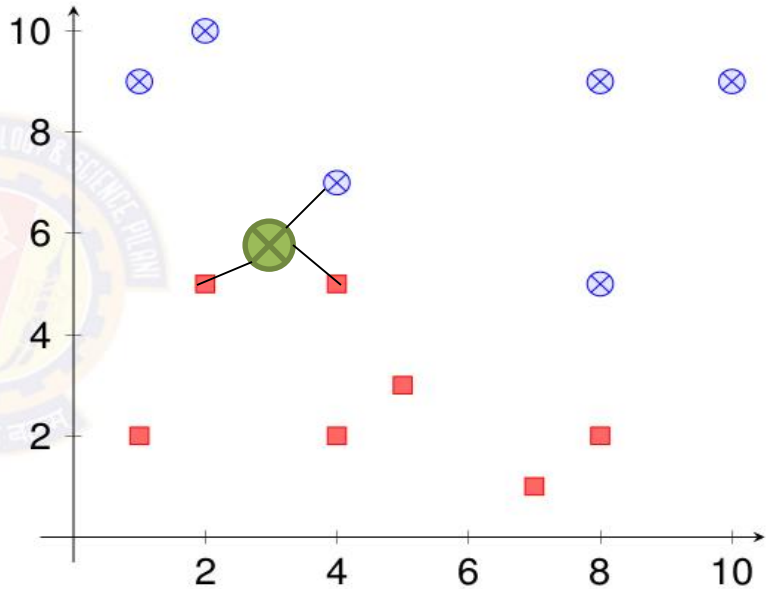


Instance Based Learning

Nearest Neighbor Approach : Notations

- K-nearest neighbor:
- Given x_q, take **vote** among its k nearest neighbors (if **discrete-valued target function**)
- Take **mean(or median)** of f values of k nearest neighbors (if **real-valued**) $f^*(x_q)=\Sigma_{i=1}^{k} f(x_i)/k$

x_1	x_2	y
1	9	white
10	9	white
4	7	white
4	5	pink
5	3	pink
8	9	white
4	2	pink
2	5	pink
7	1	pink
2	10	white
8	5	white
1	2	pink
8	2	pink

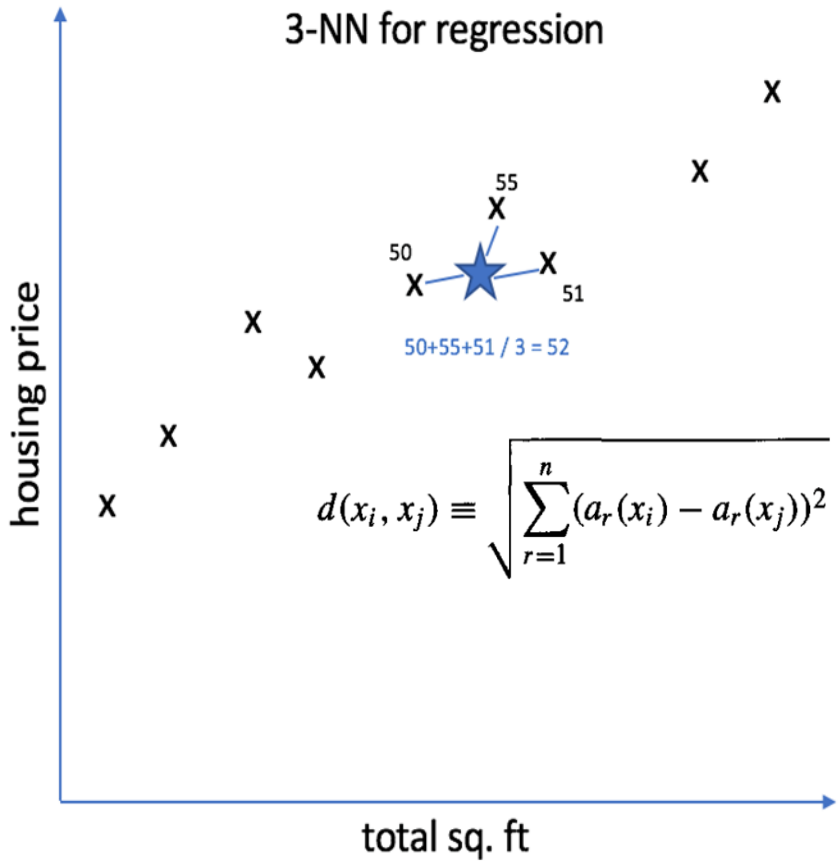


Interpretation : Assume that the Instances represented as points in a Euclidean space

Instance Based Learning

K-Nearest Neighbor Regression: Example

Size in feet^2 (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
1852	178
…	…



$$d(x_i, x_j) \equiv \sqrt{\sum_{r=1}^{n} (a_r(x_i) - a_r(x_j))^2}$$

Source Credit : https://www.jeremyjordan.me/k-nearest-neighbors/

• Measures of Dis/similarity:

- Data matrix
  - n data points with p dimensions

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix
  - n data points, but registers only the distance
  - A triangular matrix

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \cdots & \cdots & 0 \end{bmatrix}$$

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- Measures of Dis/similarity:
- Nominal Attributes

- Can take 2 or more states, e.g., red, yellow, blue, green (generalization of a binary attribute)
- <u>Method 1</u>: Simple matching
  - *m*: # of matches, *p*: total # of variables $d(i, j) = \frac{p - m}{p}$
- <u>Method 2</u>: Use a large number of binary attributes
  - creating a new binary attribute for each of the *M* nominal states

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• Measures of Dis/similarity:
• Binary Attributes

- A contingency table for binary data

		Object *j*		
		1	0	sum
Object *i*	1	q	r	$q+r$
	0	s	t	$s+t$
	sum	$q+s$	$r+t$	p

- Distance measure for symmetric binary variables:

$$d(i,j) = \frac{r+s}{q+r+s+t}$$

- Distance measure for asymmetric binary variables:

$$d(i,j) = \frac{r+s}{q+r+s}$$

- Jaccard coefficient (*similarity* measure for *asymmetric* binary variables):

$$sim_{Jaccard}(i,j) = \frac{q}{q+r+s}$$

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• Measures of Dis/similarity:
• Binary Attributes

- Example

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

  - Gender is a symmetric attribute
  - The remaining attributes are asymmetric binary
  - Let the values Y and P be 1, and the value N is 0

$$d(i, j) = \frac{r + s}{q + r + s}$$

$$d(jack, mary) = \frac{0+1}{2+0+1} = 0.33$$

$$d(jack, jim) = \frac{1+1}{1+1+1} = 0.67$$

$$d(jim, mary) = \frac{1+2}{1+1+2} = 0.75$$

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Measures of Dis/similarity

Distances: Minkowski Distance

- *Minkowski distance*: A popular distance measure

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \ldots, x_{jp})$ are two p-dimensional data objects, and h is the order (the distance so defined is also called L-h norm)

- Properties
  - $d(i, j) > 0$ if $i \neq j$, and $d(i, i) = 0$ (Positive definiteness)
  - $d(i, j) = d(j, i)$ (Symmetry)
  - $d(i, j) \leq d(i, k) + d(k, j)$ (Triangle Inequality)
- A distance that satisfies these properties is a metric

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Measures of Dis/similarity

Distances : Special Cases of Minkowski

- $h = 1$: Manhattan (city block, L_1 norm) distance
  - E.g., the Hamming distance: the number of bits that are different between two binary vectors

$$d(i,j) = |x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}| + \ldots + |x_{i_p} - x_{j_p}|$$

- $h = 2$: (L_2 norm) Euclidean distance

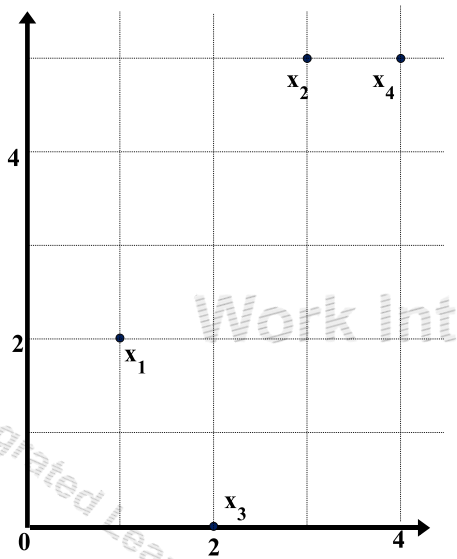
$$d(i,j) = \sqrt{(|x_{i_1} - x_{j_1}|^2 + |x_{i_2} - x_{j_2}|^2 + \ldots + |x_{i_p} - x_{j_p}|^2)}$$

- $h \to \infty$. "supremum" (L_{max} norm, L_∞ norm) distance.
  - This is the l the vectors

$$d(i,j) = \lim_{h \to \infty} \left(\sum_{f=1}^{p} |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_{f}^{p} |x_{if} - x_{jf}|$$

Example: Minkowski Distance

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



(Dissimilarity Matrices)

Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Euclidean (L_2)

L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

Supremum

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

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Standardizing Numeric Data

- Z-score:
  - X: raw score to be standardized, μ: mean of the population, σ: standard deviation
  - the distance between the raw score and the population mean in units of the standard deviation
  - negative when the raw score is below the mean, "+" when above

Where

$$z = \frac{x-\mu}{\sigma}$$

Work Integrated Learning Programmes

Measures of Dis/similarity : Ordinal Variables

- An ordinal variable can be discrete or continuous
- Order is important, e.g., rank $r_{if} \in \{1, \ldots, M_f\}$
- Can be treated like interval-scaled $z_{if} = \frac{r_{if} - 1}{M_f - 1}$
  - replace x_{if} by their rank
  - map the range of each variable onto [0, 1] by replacing *i*-th object in the *f*-th variable by

Measures of Dis/similarity
Attributes of Mixed Type: Gower distance

- A database may contain all attribute types
  - Nominal, symmetric binary, asymmetric binary, numeric, ordinal

$$d(i,j) = \frac{\sum_{c=1}^{n} \omega_c \delta_{ij}^{(c)} d_{ij}^{(c)}}{\sum_{c=1}^{n} \omega_c \delta_{ij}^{(c)}}$$

$$\begin{aligned}
d(i,j) &= \text{dissimilarity between row i and row j} \\
c &= \text{the cth column} \\
n &= \text{number of columns in the dataset} \\
\omega_c &= \text{weight of cth column} = \frac{1}{\text{nrows in dataset}} \\
\delta_{ij}^{c} &= \begin{cases} 0 & \text{if column c is missing in row i or j} \\ 0 & \text{if column c is asymmetric binary and both values in row i and j are 0} \\ 1 & \text{otherwise} \end{cases} \\
d_{ij}^{c}(\text{categorical}) &= \begin{cases} 0 & \text{if i and j are equal in column c} \\ 1 & \text{otherwise} \end{cases} \\
d_{ij}^{c}(\text{continuous/ordinal}) &= \frac{|\text{row i in column c - row j in column c}|}{\text{max(column c) - min(column c)}}
\end{aligned}$$

https://healthcare.ai/clustering-non-continuous-variables/

Example

Based on the information given in the table below, find most similar and most dissimilar persons among them. Apply min-max normalization on income to obtain [0,1] range. Consider profession and mother tongue as nominal. Consider native place as ordinal variable with ranking order of [Village, Small Town, Suburban, Metropolitan]. Give equal weight to each attribute.

Name	Income	Profession	Mother tongue	Native Place
Ram	70000	Doctor	Bengali	Village
Balram	50000	Data Scientist	Hindi	Small Town
Bharat	60000	Carpenter	Hindi	Suburban
Kishan	80000	Doctor	Bhojpuri	Metropolitan

Solution

After normalizing income and quantifying native place, we get

Name	Income	Profession	Mother tongue	Native Place
Ram	0.67	Doctor	Bengali	1
Balram	0	Data Scientist	Hindi	2
Bharat	0.33	Carpenter	Hindi	3
Kishan	1	Doctor	Bhojpuri	4

d(Ram, Balram) = 0.67+1+1+(2-1)/(4-1)=3 d(Ram, Bharat) = 0.33+1+1+(3-1)/(4-1)=3
d(Ram, Kishan) = 0.33+0+1+(4-1)/(4-1) = 2.33 d(Balram, Bharat) = 0.33+1+0+(3-2)/(4-1)=1.67
d(Balram, Kishan) = 1+1+1+(4-2)/(4-1) = 3.67 d(Bharat, Kishan) = 0.67+1+1+(4-3)/(4-1) = 3

Most similar – Balram and Bharat; Most dissimilar – Balram and Kishan

Gower's Distance = d(i,j) / 4 (with equal weights it becomes simple average)
It will range from 0 to 1.

Measures of Dis/similarity

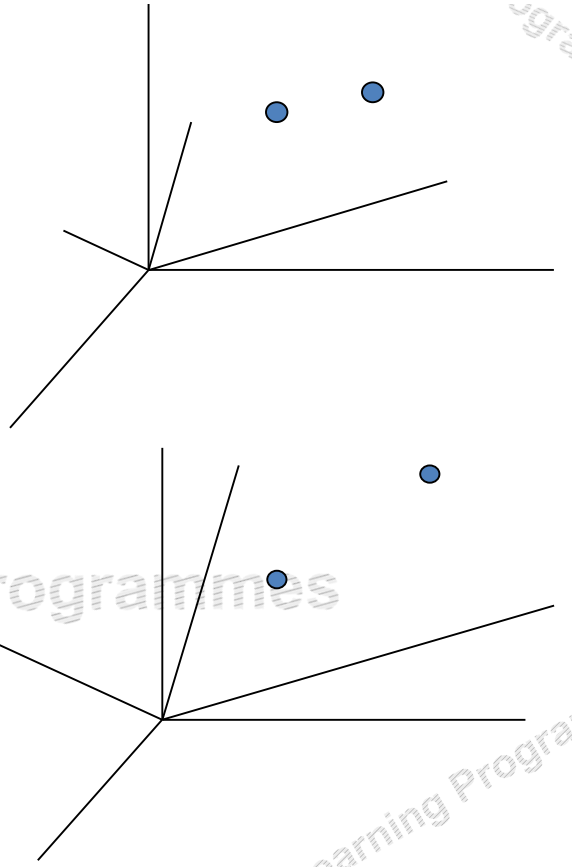
Curse of Dimensionality

- It uses all attributes to classify instances
- If the target concept depends on only a few of the many available attributes, then the instances that are truly most "similar" may well be a large distance apart.
- ***Curse of dimensionality***: nearest neighbor is easily misled when instance space is high-dimensional

  Solution 1: weigh the attributes differently (use cross-validation to determine the weights)

  Solution 2: eliminate the least relevant attributes (again, use cross-validation to determine which attributes to eliminate)

Stretching the axes in the Euclidean space, shortening the axes that correspond to less relevant attributes, and lengthening the axes that correspond to more relevant attributes



Instance Based Learning

K-Nearest Neighbor : Algorithm

Approximating a discrete-valued function $f : \Re^n \rightarrow V$

Training algorithm:

- For each training example $\langle x, f(x) \rangle$, add the example to the list *training_examples*

Classification algorithm:

- Given a query instance x_q to be classified,
  - Let $x_1 \ldots x_k$ denote the k instances from *training_examples* that are nearest to x_q
  - Return

$$\hat{f}(x_q) \leftarrow \underset{v \in V}{\operatorname{argmax}} \sum_{i=1}^{k} \delta(v, f(x_i))$$

where $\delta(a, b) = 1$ if $a = b$ and where $\delta(a, b) = 0$ otherwise.

Approximate a real-valued target function $f : \Re^n \rightarrow \Re$

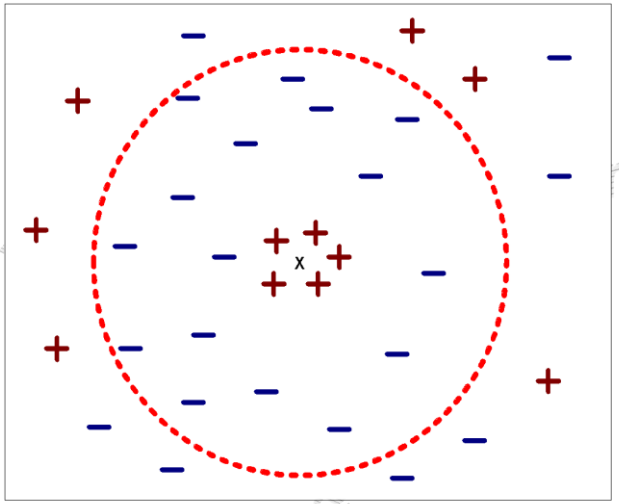
$$\hat{f}(x_q) \leftarrow \frac{\sum_{i=1}^{k} f(x_i)}{k}$$

Instance Based Learning

K-Nearest Neighbor : Hyper parameter “K”

Choosing the value of k:

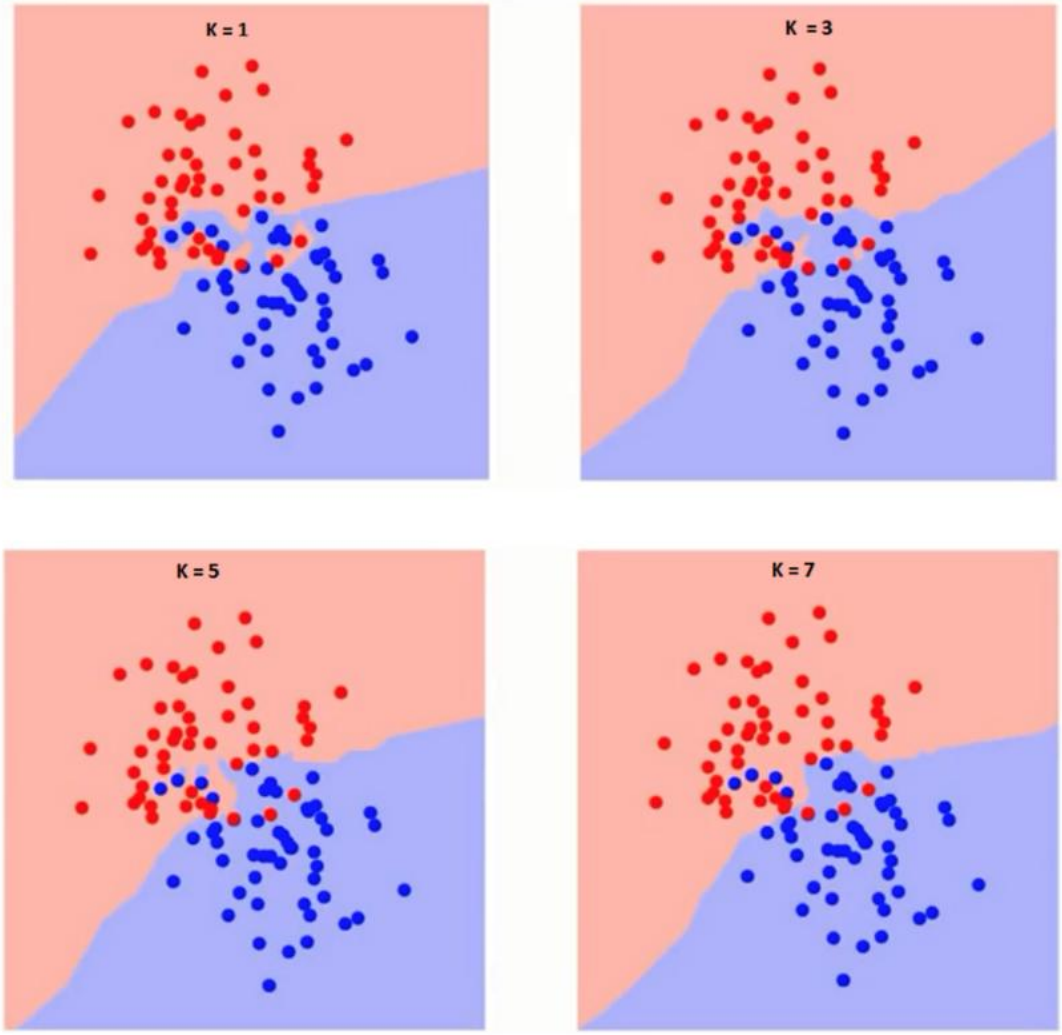
Rule of thumb:
K = sqrt(N)
N: number of training points



Instance Based Learning

K-Nearest Neighbor : Hyper parameter “K” : Elbow Method

- Compute sum of squares error (SSE) or any other error function for varying values of K (1 to a reasonable X) and plot against K
- In the plot, the elbow (see pic) gives the value of K beyond which the error function plot almost flattens
- As K approaches the total number of instances in the set, error function drops down to ‘0’



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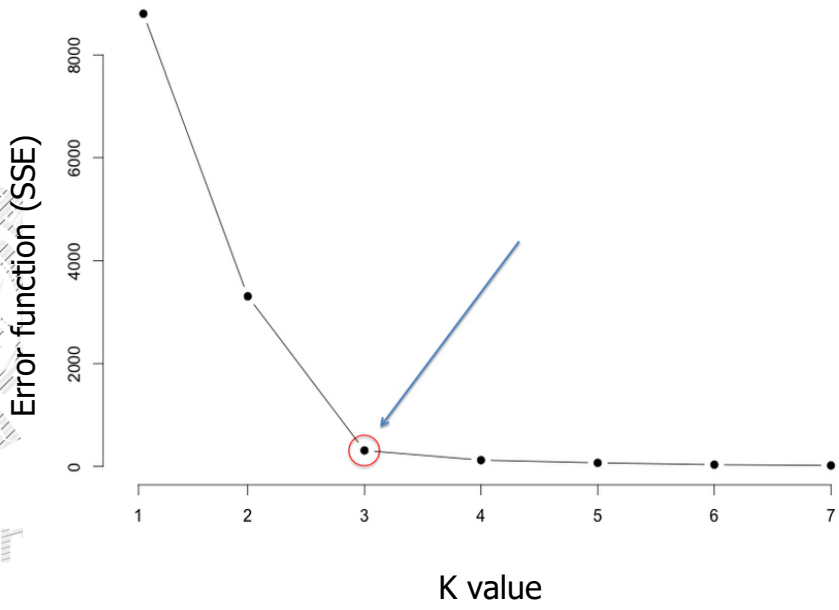
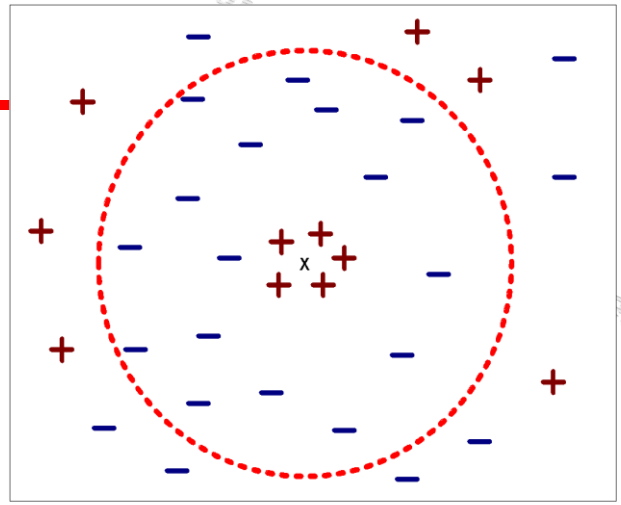


Instance Based Learning

K-Nearest Neighbor : Variation



- Distance Weighted K-NN algorithm

contribution of each of the k nearest neighbors is weighted accorded to their distance to x_q

- **discrete-valued target functions**

$$\hat{f}(x_q) \leftarrow \underset{v \in V}{argmax} \sum_{i=1}^{k} w_i \delta(v, f(x_i))$$

where $w_i \equiv \frac{1}{d(x_q, x_i)^2}$ and $\hat{f}(x_q) = f(x_i)$ if $x_q = x_i$

- **continuous-valued target function:**

$$\hat{f}(x_q) \leftarrow \frac{\sum_{i=1}^{k} w_i f(x_i)}{\sum_{i=1}^{k} w_i}$$

<u>Kernel Functions</u>

$w_i = K(d(x_q, x_i))$

$K(d(x_q, x_i)) = 1/\, d(x_q, x_i)^2$

$K(d(x_q, x_i)) = 1/(d_0 + d(x_q, x_i))^2$

$K(d(x_q, x_i)) = \exp(-(d(x_q, x_i)/\sigma_0)^2)$

$d(x_q, x_i)$ is the distance between x_q and x_i

Problem Type – 1

Consider the following training set in 2-dimensional Euclidean space. What is the class of the point (1, 1) if 7NN classifier is considered? If the value of K is reduced whether the class will change? (Consider K=3 and K=5). What should be the final class if the above 3 values of K are considered?

Point	Coordinate	Class
X1	(-1, 1)	Negative
X2	(0, 1)	Positive
X3	(0, 2)	Negative
X4	(1, -1)	Negative
X5	(1, 0)	Positive
X6	(1, 2)	Positive
X7	(2, 2)	Negative
X8	(2, 3)	Positive

Distance from (1,1)
2
1
1.414
2
1
1
1.41
2.236

Point		Class
X2	1	Positive
X5	1	Positive
X6	1	Positive
X7	1.41	Negative
X3	1.414	Negative
X1	2	Negative
X4	2	Negative

Problem Type – 2

Based on the information given in the table below, find the customer type of x_q=C4 using K-NN. In given example consider all the instances and use locally weighted K-NN algorithm with kernel $K(d(x_q,x_i)) = 1/ d(x_q,x_i)^2$.

Apply min-max normalization on income to obtain [0,1] range. Consider profession and Region as nominal. Consider :Locality as ordinal variable with ranking order of [Village, Small Town, Suburban, Metropolitan]. Give equal weight to each attribute.

Customer	Income	Profession	Region	Locality	Category
C0	60000	Doctor	Hindi	Village	L1
C1	70000	Doctor	Bengali	Village	L2
C2	60000	Carpenter	Hindi	Suburban	L2
C3	80000	Doctor	Bhojpuri	Metropolitan	L2
C5	80000	Data Scientist	Hindi	Small Town	L1
C4	50000	Data Scientist	Hindi	Small Town	

Problem Type – 2

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Customer	Income	Profession	Region	Scaled Locality	Category	Distance	
C0	0.33	Doctor	Hindi	1 □ 0	L1	**0.97**	**1.06**
C1	0.67	Doctor	Bengali	1 □ 0	L2	**1.75**	**0.33**
C2	0.33	Carpenter	Hindi	3 □ 0.67	L2	**0.97**	**1.06**
C3	1	Doctor	Bhojpuri	4 □ 1	L2	**2.2**	**0.21**
C5	1	Data Scientist	Hindi	2 □ 0.33	L1	**1**	**1**
C4	0	Data Scientist	Hindi	2 □ 0.33		0 (NA)	

d(C4, C2)

= distance w.r.t to numerical attributes + distance w.r.t Nominal attributes

= Euclidean distance on (Income, Scaled Locality) + Categorical distance on attributes (Profession, Region)

$= \sqrt{((Income_{c4}-Income_{c2})^{"2"} + (Locality_{c4}-Locaility_{c2})^2)} + (No.of.Categorical\ features - No.of.Matches\ between\ C4\ \&\ C2)/(No.of.Categorical\ Features)$

$= \sqrt{((0-0.33)^{"2"} + (0.33-0.67)^2)} + (2-1)/2$

= 0.47 + 0.5 = 0.97

Problem Type – 2

Based on the information given in the table below, find the customer type of x_q=C4 using K-NN. In given example consider all the instances and use locally weighted K-NN algorithm with kernel $K(d(x_q,x_i)) = 1/ d(x_q,x_i)^2$.

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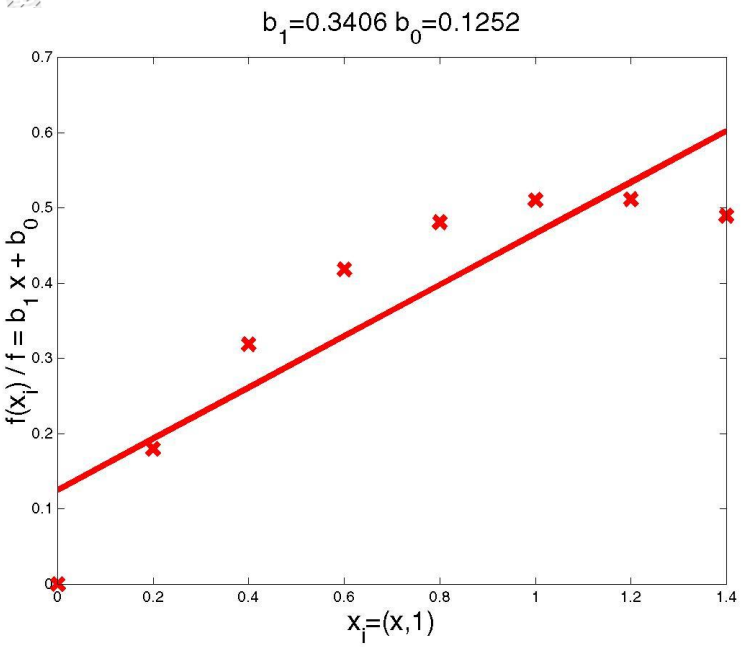
C1	0.67	Doctor	Bengali	1 □ 0	L2	**1.75**	**0.33**
C2	0.33	Carpenter	Hindi	3 □ 0.67	L2	**0.97**	**1.06**
C3	1	Doctor	Bhojpuri	4 □ 1	L2	**2.2**	**0.21**
C5	1	Data Scientist	Hindi	2 □ 0.33	L1	**1**	**1**
C4	0	Data Scientist	Hindi	2 □ 0.33		0 (NA)	

Point	Kernel / Weight	Class
C5	1	L1
C0	1.06	L1
C2	1.06	L2
C1	0.33	L2
C3	0.21	L2

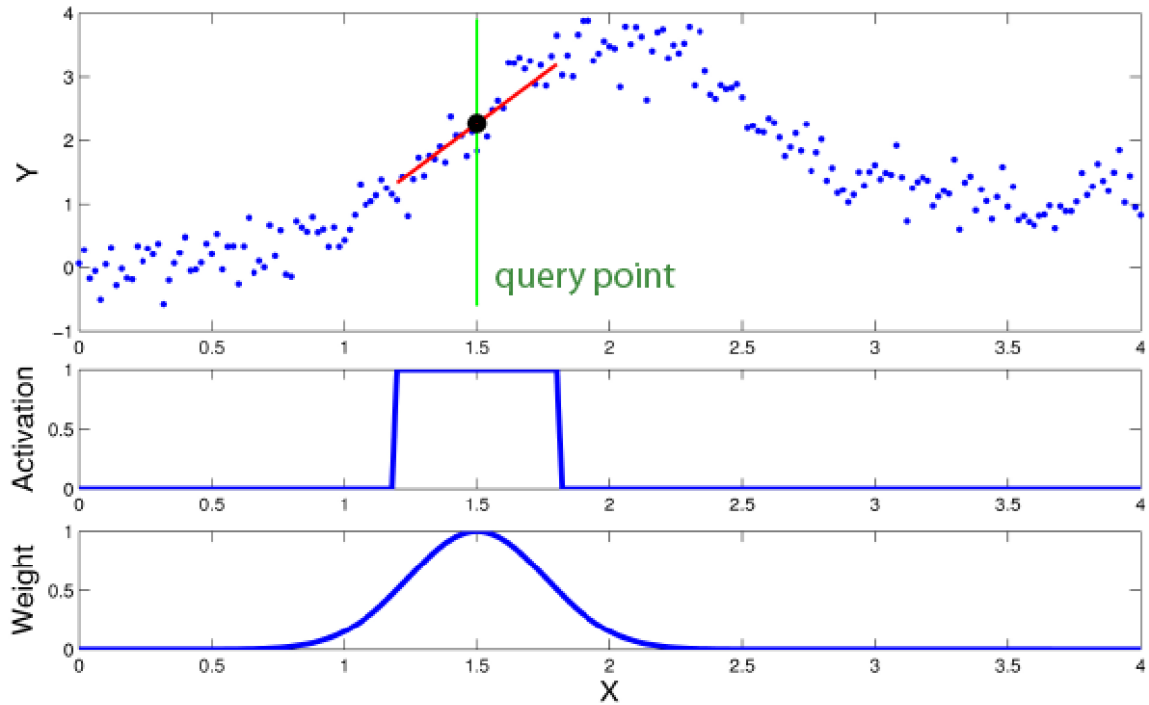
Inference : Without weighted KNN, L2 is the majority voted class.
But with weighted distances, L1 class instances are more similar to C4

Locally Weighted Regression



Locally Weighted Regression

- Locally – Function approximated based on data near query point
- Weighted – Contribution by each training example is weighted by its distance from query point
- Regression- Approximates real-valued target function
- ***Residual*** is the error in approximating the target function. $\hat{f}(x) - f(x)$
- ***Kernel function*** is the function of distance that is used to determine the weight of each training example. In other words, the kernel function is the function ***K*** such that $w_i = K(d(x_i, x_q))$



Source Credit : https://www.ias.informatik.tu-darmstadt.de/uploads/Teaching/AutonomousLearningSystems/Englert_ALS_2012.pdf

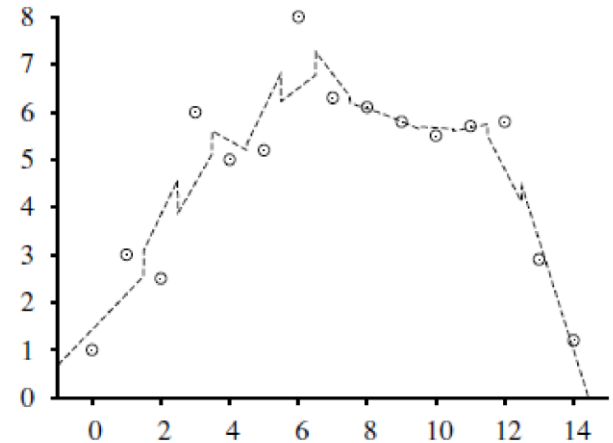
Locally Weighted Regression

- The nearest-neighbor approaches can be thought of as approximating the target function f ***(x)*** at the single query point ***x*** = x_q

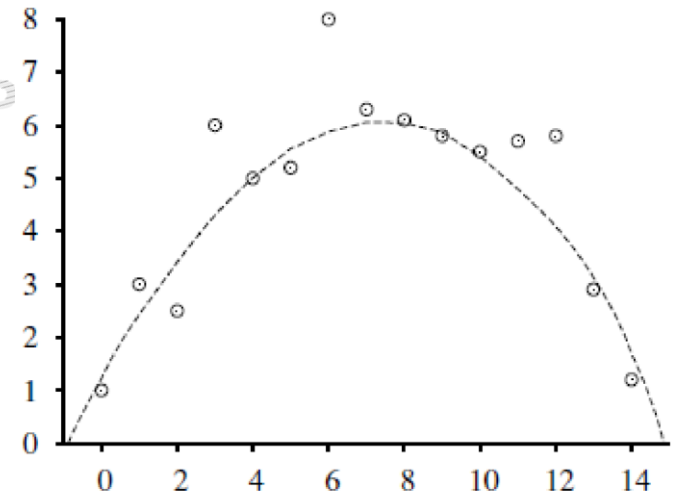
- Locally weighted regression is a generalization of this approach. It constructs an explicit approximation to f over a local region surrounding x_q

- Approximate the target function in the neighborhood surrounding x, using a linear function, a quadratic function, a multilayer neural network, or some other functional form.

3-nearest-neighbors linear regression



Locally weighted regression with a quadratic kernel



Locally weighted linear regression

- target function is approximated using a **linear function**

$$\hat{f}(x) = w_0 + w_1 a_1(x) + ... + w_n a_n(x)$$

- methods like **gradient descent** can be used to calculate the coefficients $w_0, w_1, ..., w_n$ to minimize the error in fitting such linear functions

- ANNs require a global approximation to the target function
- here, just a local approximation is needed

$\Rightarrow$ the error function has to be redefined

Locally weighted linear regression

- possibilities to redefine the error criterion E
  1. Minimize the squared error over just the k nearest neighbors

$$E_1(x_q) \equiv \frac{1}{2} \sum_{x \in k \text{ nearest neigbors}} (f(x) - \hat{f}(x))^2$$

  2. Minimize the squared error over the entire set D, while weighting the error of each training example by some decreasing function K of its distance from x_q

$$E_2(x_q) \equiv \frac{1}{2} \sum_{x \in D} (f(x) - \hat{f}(x))^2 \cdot K(d(x_q, x))$$

  3. Combine 1 and 2

$$E_3(x_q) \equiv \frac{1}{2} \sum_{x \in k \text{ nearest neighbors}} (f(x) - \hat{f}(x))^2 \cdot K(d(x_q, x))$$

Locally weighted linear regression

- For a given query point $\mathbf{x}_q$ we solve the following weighted regression problem using gradient descent training rule:

$$\Delta w_j = \eta \sum_{x \in k \text{ nearest nbrs of } x_q} K(d(x_q, x))\ (f(x) - \hat{f}(x))\ a_j(x)$$

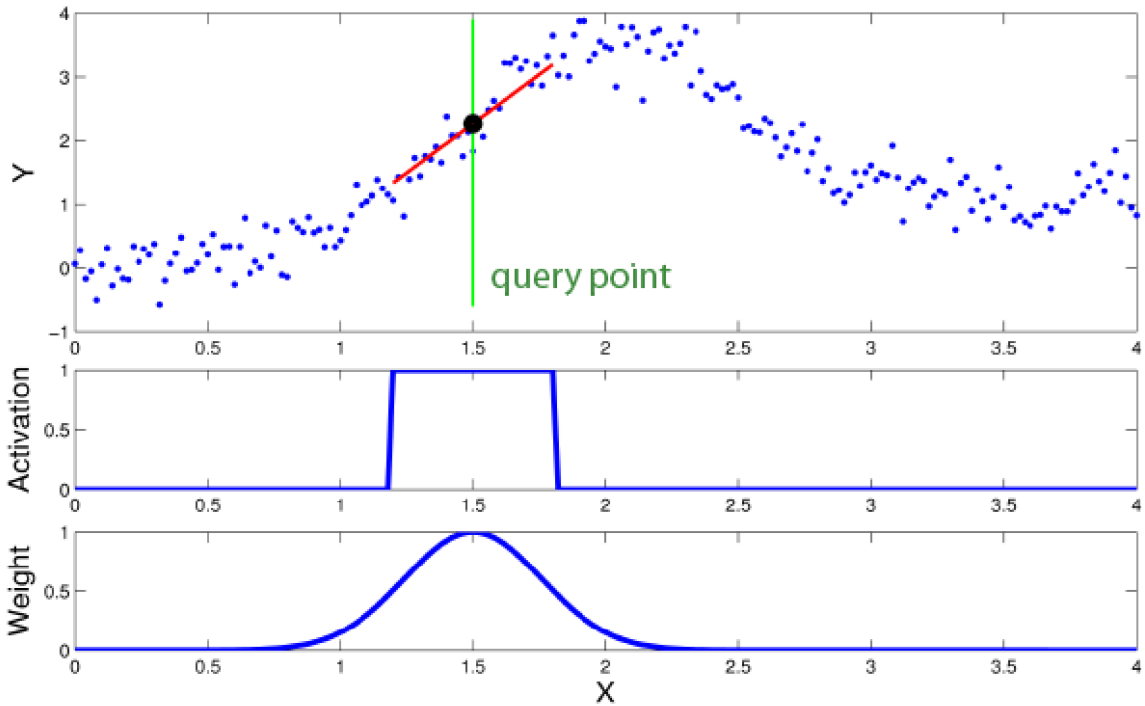
- Note that we need to solve a new regression problem for *every* query point—that's what it means to be *local*.
- In ordinary linear regression, we solved the regression problem once, globally, and then used the same h_w for any query point.

$$w_i \leftarrow w_i + \Delta w_i$$

Note: + sign in weight update rule as in Δw_i (actual-prediction) is used.

If we use (prediction – actual) then instead of '+', '-' sign will be used.

Example



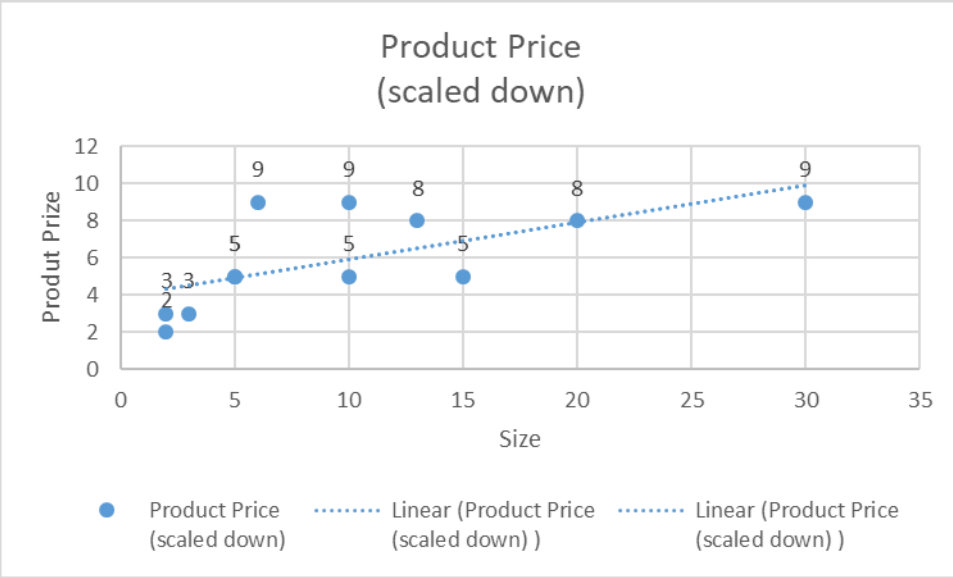
- The upper graphic the set of data points (x,y) (blue dots), query point (green line), local linear model (red line) and prediction (black dot).
- The graphic in the middle shows the activation area of the model.
- The corresponding weighting kernel (receptive field) is shown in the bottom graphic.

Source Credit : https://www.ias.informatik.tu-darmstadt.de/uploads/Teaching/AutonomousLearningSystems/Englert_ALS_2012.pdf

Problem Type - 3

Consider the following training set in 2-dimensional Euclidean space. What is the prize of the product with size 7 if 5-NN is considered? Use the kernel of the form $K(d(x_q,x_i)) = \exp(- d(x_q,x_i)^2 / 2b^2)$ where b=1. Use the linear regression of the form y = 3.5 + 0.8 x and the learning rate of 0.2 to apply the gradient descent and update the weights at the end of first iteration

Size	Product Price (scaled down)
10	5
5	5
2	2
3	3
5	5
15	5
30	9
20	8
6	9
2	3
10	9
13	8



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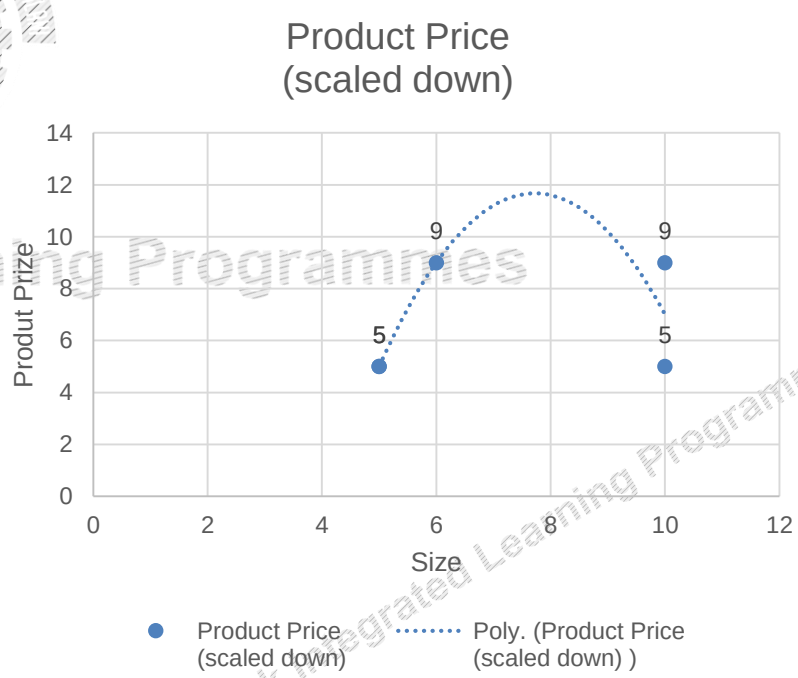


Problem Type - 3

Use either of below formula :

$$W_{j-new} = W_{j-old} + \alpha * \sum_{i=1}^{K\ Neighbors} K(d(x_q,x_i))(f(x_i) - f\hat{}(x_i))a_j(x_i)$$

$$W_{j-new} = W_{j-old} - \alpha * \sum_{i=1}^{K\ Neighbors} K(d(xq,xi))(f\hat{}(x_i) - f(x_i))a_j(x_i)$$

Size	Product Price (scaled down)		
10	5	3	0.01
5	5	2	0.14
2	2	5	0
3	3	4	0
5	5	2	0.14
15	5	8	0
30	9	23	0
20	8	13	0
6	9	1	0.61
2	3	5	0
10	9	3	0.01
13	8	6	0

Apply gradient descent where the gradient is weighed by the kernel output.

Answer :

Updated weights after 1st iteration

w0 = 3.43, w1= 0.43

Price Prediction	
6.459515	After 1st iteration of GD
9.1	Before GD

When to Consider Nearest Neighbors

- Suitable for Low dimensional datasets
- Local information can yield highly adaptive behavior
- Lots of training data (distance-weighted KNN)
    - Training is very fast
- Learn complex target functions
    - Do not lose information
- Noisy training data (distance-weighted KNN)
    - by taking the weighted average of the k neighbors nearest to the query point, it can smooth out the impact of isolated noisy training examples.

Nearest-Neighbor Classifiers: Challenges

- The value of *k*, the number of nearest neighbors to retrieve
- Choice of Distance Metric to compute distance between records
- Expensive, Slow at query time
  - To determine the nearest neighbour of a query point *q*, must compute the distance to all *N* training examples
    - + Pre-sort training examples into fast data structures (kd-trees)
    - + Remove redundant data (condensing)
- Storage Requirements
  - Must store all training data **P**
    - + Remove redundant data (condensing)

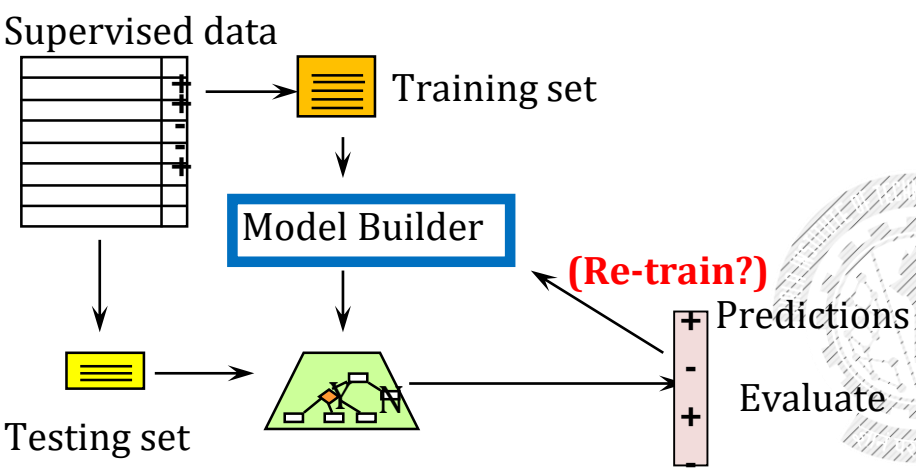
Evaluating Classifier Performance

Work Integrated Learning Programmes

Model selection and assessment

- Model selection: estimating the performance of different models in order to choose the best one.
- Model assessment: having chosen a final model, estimating its prediction error (generalization error) on new data.
- To estimate generalization error (measure the quality of inductive bias), we need data unseen during training. We split the data as
  - Training set : used to fit the models
  - Validation set : used to estimate prediction error for model selection, tune hyperparameters, feature selection
  - Test (publication) set : is used for assessment of the generalization error of the final chosen model

Hyper parameter tuning using test set



Problem 1: Hesitating between two models (e.g linear model and a polynomial model) ?
Solution 1: train both and compare how well they generalize using the test set: linear model generalizes better but looks like overfitting So, apply some regularization
Problem 2: how to choose the value of regularization hyper parameter?
Solution 2: train 100 different models using 100 different values for this hyper parameter. Choose best hyper parameter value with the lowest generalization error (5%). Great!!: launch into production

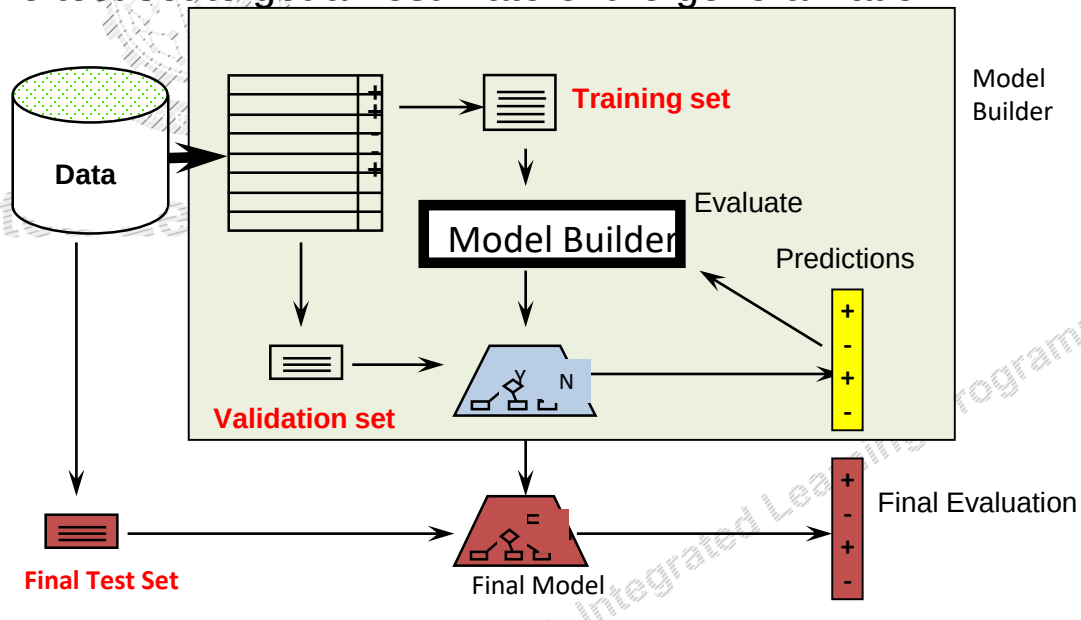
Problem 3: Error now is increased to 5% errors. Why?
Answer 3: Generalization error was measured multiple times on the test set, So model and hyper parameters are adapted to produce the best model for that particular set, unlikely to perform as well on new data.

Learning : The test data can't be used for parameter tuning!

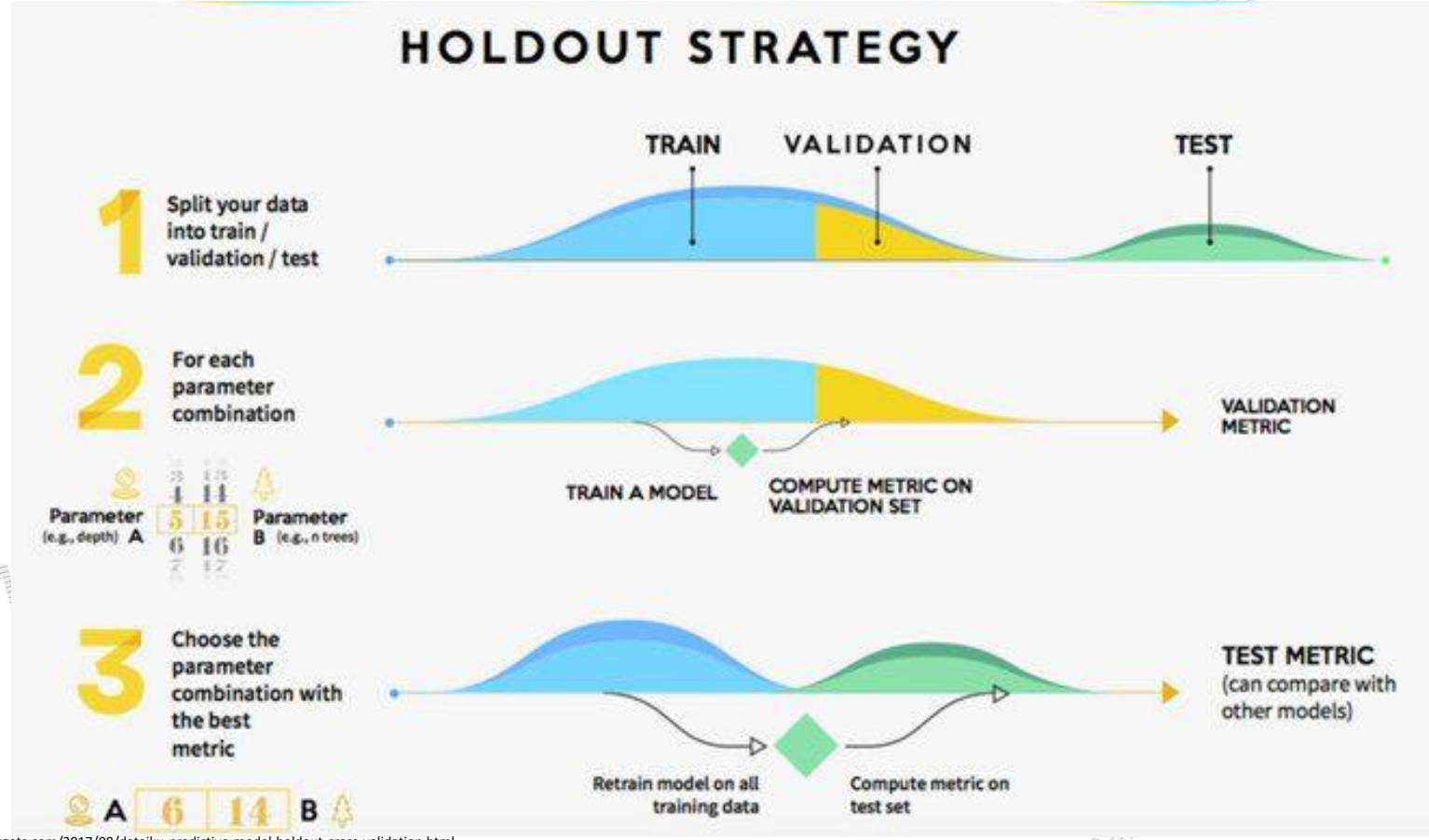
Train, Validation, Test split

Holdout validation (holdout or development, or dev set)

1. Train multiple models with various hyper parameters on the reduced training set (i.e., the full training set minus the validation set)
2. Select the model that performs best on the validation set
3. After this holdout validation process, you train the best model on the full training set (including the validation set), to get the final model.
4. Lastly, evaluate this final model on the test set to get an estimate of the generalization error



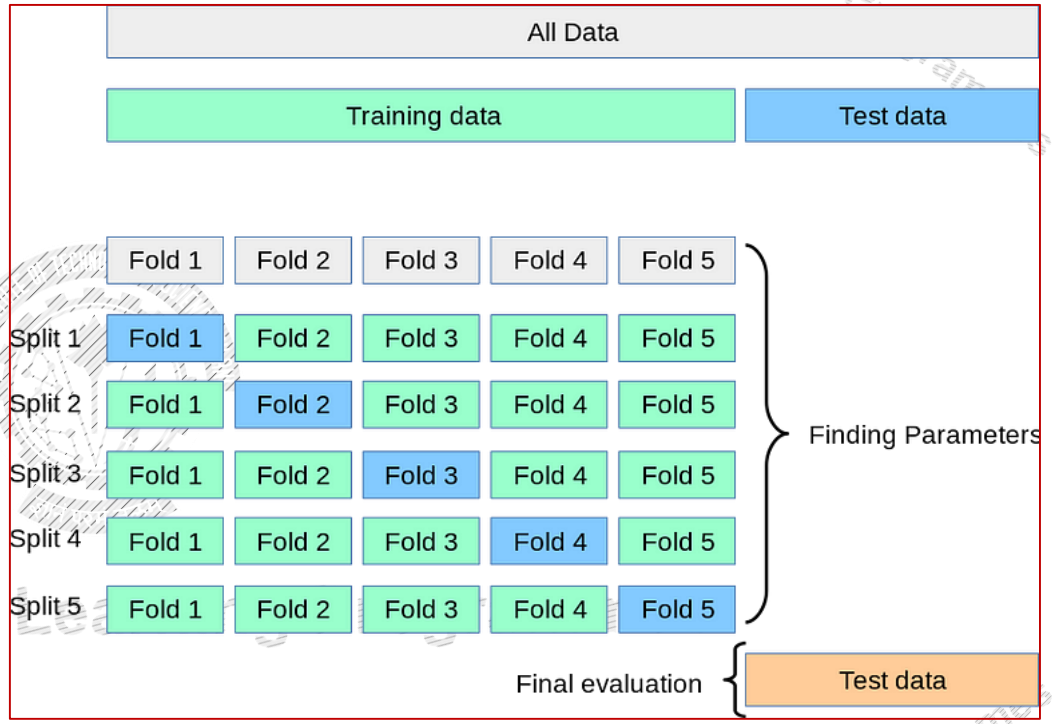
Train-Test-Validation



https://www.kdnuggets.com/2017/08/dataiku-predictive-model-holdout-cross-validation.html

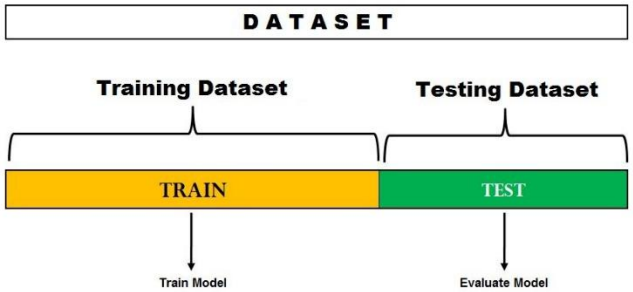
Making most out of the data

- We don't want to touch the test set until we are ready to launch a model we are confident about
- If the validation set is too small, then model evaluations will be imprecise
  - Larger the validation data the more accurate the error estimate
- If the validation set is too large, then the remaining training set will be much smaller than the full training set.
  - Larger the training data the better the classifier
- **Solution: cross validation**

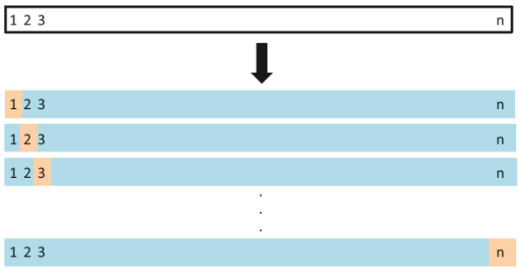


Cross-Validation techniques

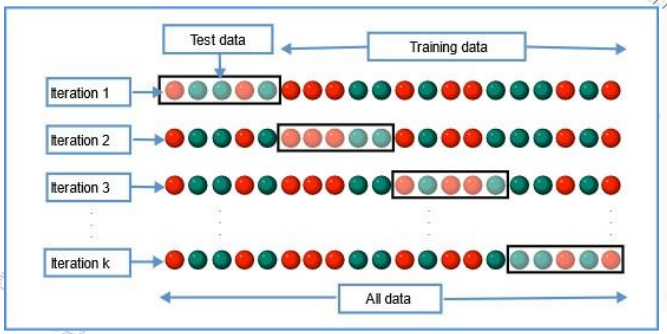
Hold Out method



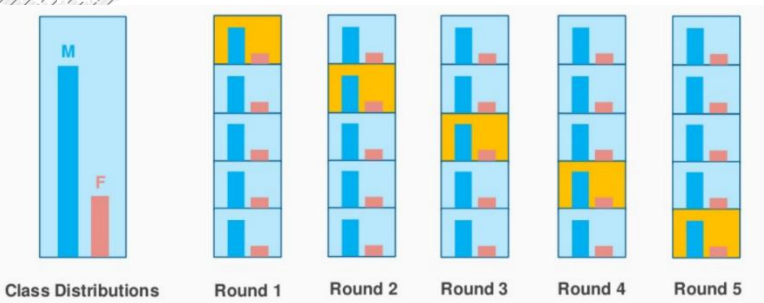
Leave One Out Cross-Validation



K-Fold Cross-Validation

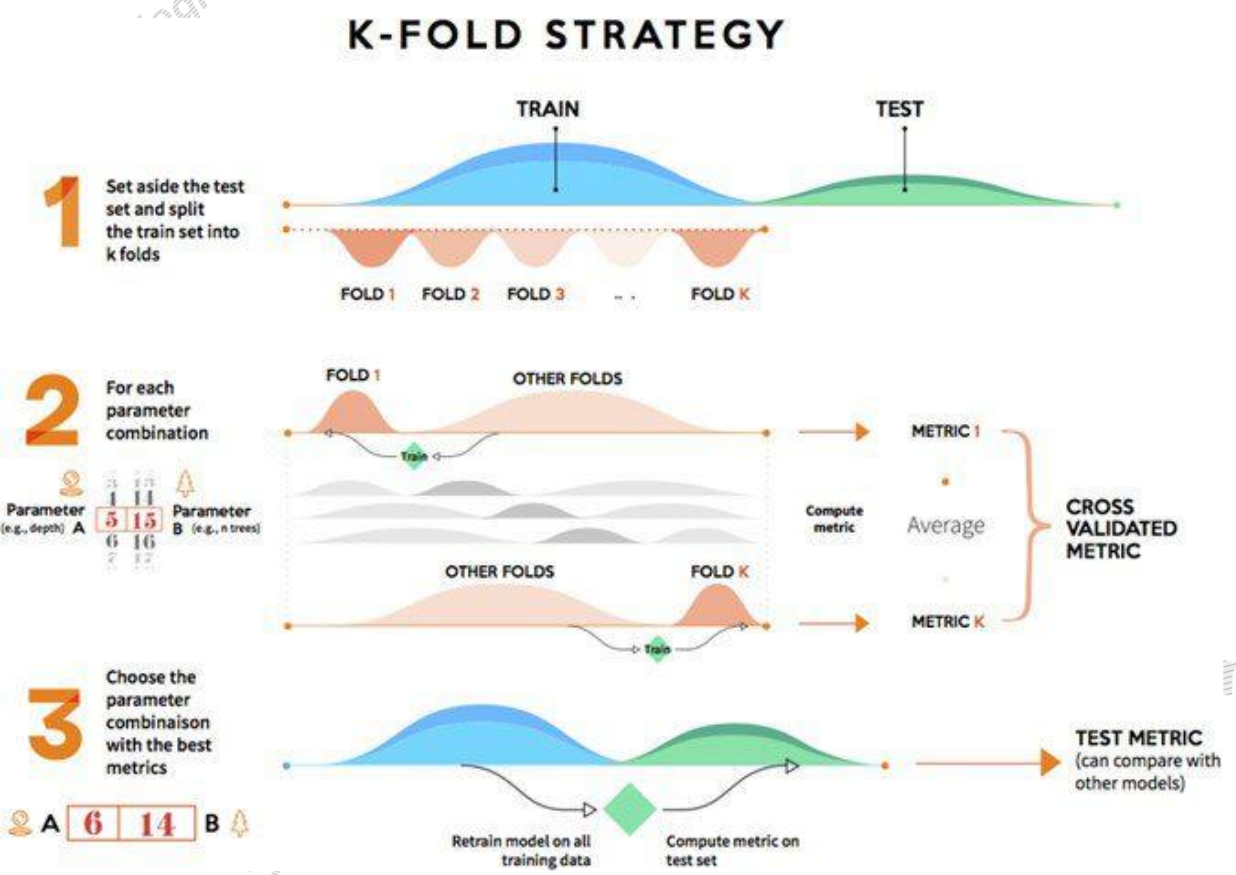


Stratified K-Fold Cross-Validation



https://www.analyticsvidhya.com/blog/2021/05/4-ways-to-evaluate-your-machine-learning-model-cross-validation-techniques-with-python-code/

Cross-Validation for hyper parameter tuning



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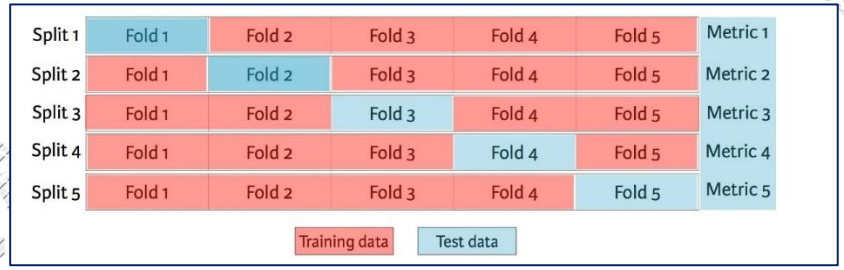
Estimating generalization error : cross validation set Vs test set

- Training set performance is not a good indicator of generalization error
  - A more complex model overfits, a less complex one underfits
  - Which model do I select?
- Test set
  - Typically 80%, 20%
  - Wastes valuable labeled data
- Cross validation
  - In Sklearn : cross_val_score() (default is accuracy)
  - cross_val_score will automatically split data into train and test
  - fit the model on train data
  - score on test data
  - For 10 folds, 10 scores will be returned in scores average of that




```
model=DecisionTreeClassifier()
scores = cross_val_score(model, X, y, cv=10)
avg_score = np.mean(score_array)
print(avg_score)
```